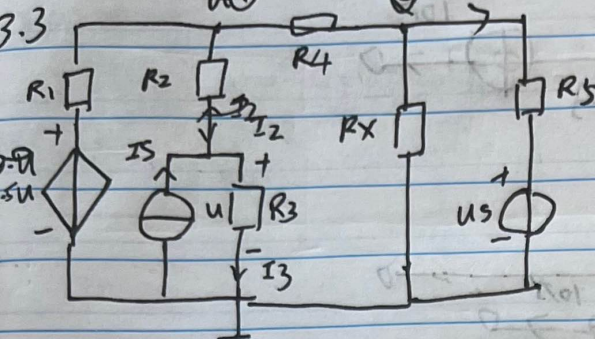


第3次作业



$\therefore US$ 中电流为 0

设节点 ① 电压为 U_1

$\therefore$ 有 $\frac{1}{R_1} + \frac{1}{R_2+R_3} + \frac{1}{R_4+R_X} U_1 = \frac{3.5U}{R_1} + \frac{I_S R_3}{R_2+R_3}$

有 $U_1 = I_2 R_2 + I_3 R_3$

$I_S + I_2 = I_3$

$U = I_3 R_3$

$\therefore US$ 中电流为 0

$\therefore RX$ 两端电压 $U_2 = 5V$ 有 $U_1 = \frac{R_4+R_X}{R_X} U_2$

联立有 $R_X = 2\Omega$ $U = 4V$

3.5 由齐性定理

$I_S = KI + b$ $I = KI_S + b$

$\therefore$ 当 $I_S = 2A$ 时 $I = -1A$

$\therefore I_S = \Rightarrow K = \frac{1}{2}$

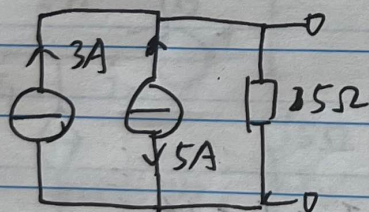
$I_S = 4A$ 时 $I = 0$

$b = -2A$

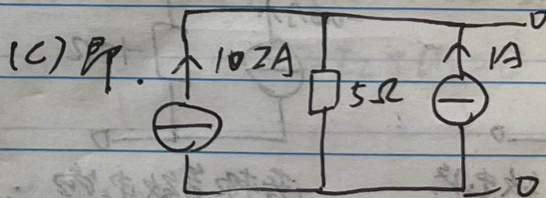
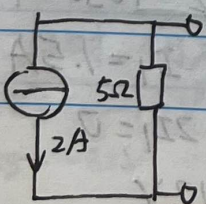
$\therefore I = \frac{1}{2} I_S - 2A$

当 $I = 1A$ 时 $I_S = 6A$

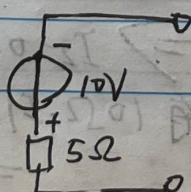
3.7 (a) 即



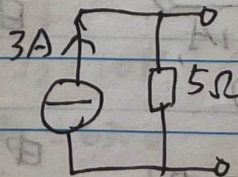
诺顿等效电路为



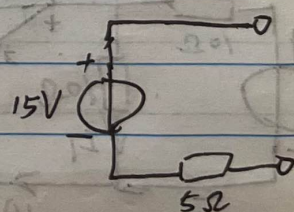
戴维南等效电路为



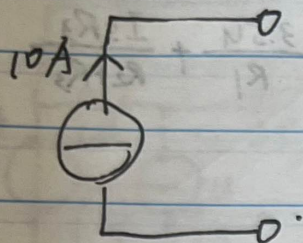
诺顿等效电路为



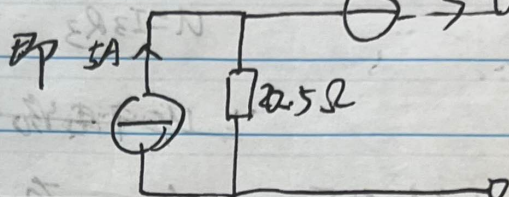
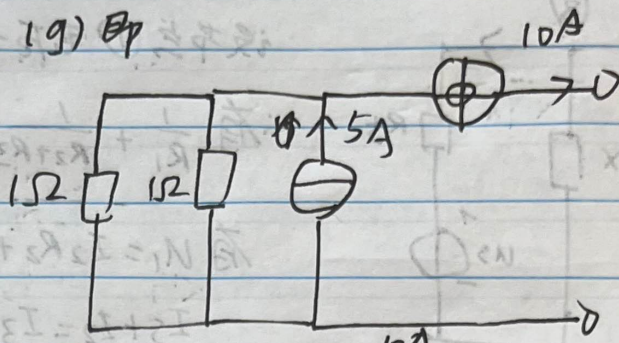
戴维南等效电路为



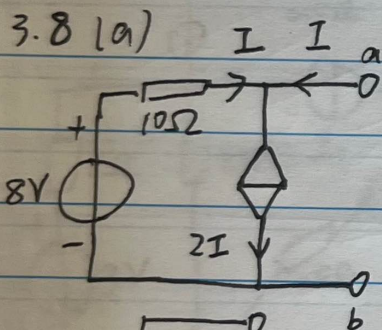
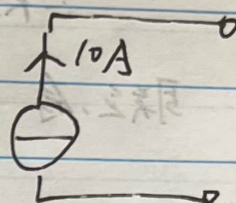
1e) 即有诺顿等效电路.



1g) 即



即只有诺顿等效电路为

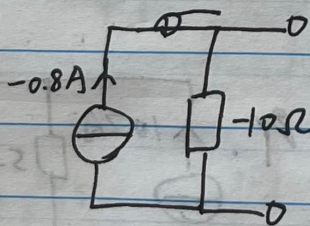
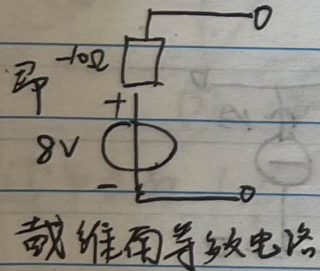


开路时, $I=0$

$$\therefore U_{ab} = 8V$$

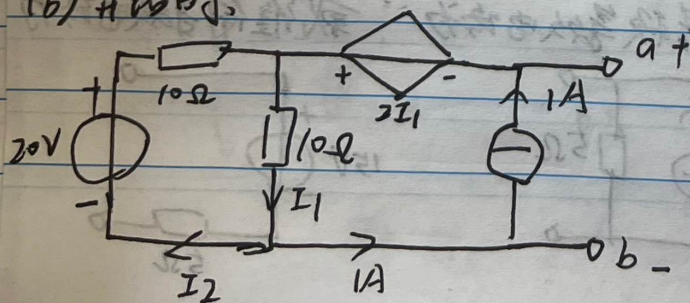
$$\text{短路时有 } I = \frac{8V}{10\Omega} = 0.8A$$

$$\therefore R_{in} = -10\Omega$$



诺顿等效电路.

1b) 开路时,



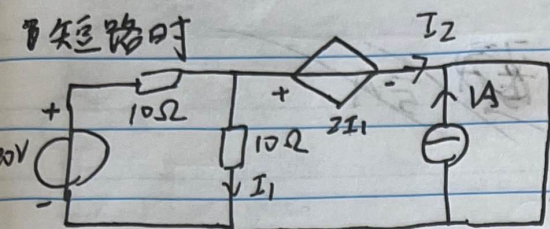
开路时有 $I_1 = I_2 + 1A$

$$\text{由 KVL, 有 } 20V - 10\Omega \cdot I_2 - 10\Omega \cdot I_1 = 0$$

$$\Rightarrow I_2 = 0.5A \quad I_1 = 1.5A$$

$$\text{由 KVL, 有 } 10\Omega \cdot I_1 - U - 2I_1 = 0$$

$$\Rightarrow U = 12V$$



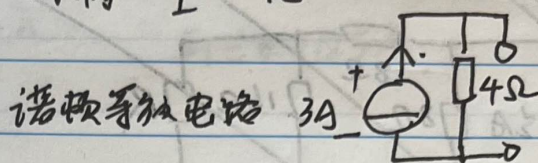
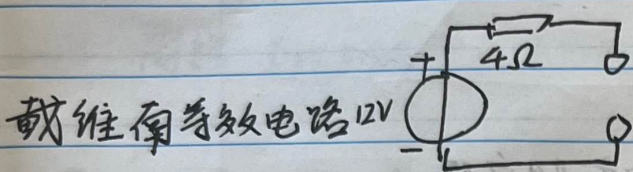
由 KVL, 有 $10\Omega \cdot I_1 - 2I_1 = 0$

$$\Rightarrow I_1 = 0 \text{ A}$$

$$\therefore I_2 = \frac{20\text{V}}{10\Omega} = 2 \text{ A}$$

$$\therefore \text{短路电流 } I = I_2 + 1 \text{ A} = 3 \text{ A}$$

$$\therefore R_{\text{内}} = \frac{U}{I} = 4\Omega$$



3.11 $\therefore R=10\Omega$ 时 $P=22.5\text{W}$, 由 $P=I^2R \Rightarrow I_1=1.5\text{A}$ $U_1=15\text{V}$

$R=20\Omega$ 时 $P=20\text{W}$ 由 $P=I^2R \Rightarrow I_2=1\text{A}$ $U_2=20\text{V}$

设由戴维南定理, 设等效电动势为 U_S , 内阻为 R_S

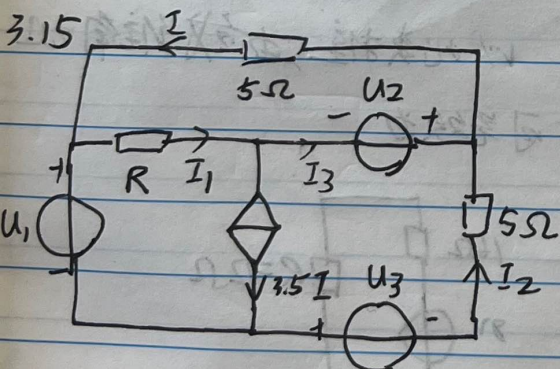
$$\therefore \text{有 } I_1 = \frac{U_S}{R_S + R_1} \quad I_2 = \frac{U_S}{R_S + R_2}$$

联立解得 $U_S = 30\text{V}$

$R_S = 10\Omega$

$$\therefore I = \frac{U_S}{R_S + 30\Omega} = 0.75\text{A}$$

$$P = I^2 R = 16.875\text{W}$$



由 KVL, 有 $5U_1 + U_2 + U_3 - I_1 R + 5\Omega \cdot I_2 = 0$

$$U_1 + 5\Omega \cdot I + 5\Omega \cdot I_2 + U_3 = 0$$

由 KCL 有 $I_1 = I_3 + 3.5I$

$$I_3 + I_2 = I \Rightarrow I = \frac{I_1 + I_2}{4.5}$$

当 $R=2\Omega$ 时, 代入有

$$U_1 + U_2 + U_3 = -10\text{V}$$

$$U_1 + U_3 = -30\text{V}$$

$$\Rightarrow \begin{cases} -I_1 R + 5\Omega \cdot I_2 = 10\text{V} \\ 5\Omega \cdot I + 5\Omega \cdot I_2 = 30\text{V} \end{cases}$$

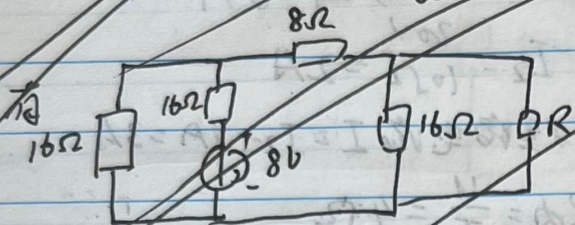
$$\Rightarrow I_1 = 2.96\text{A}$$

$$I_2 = 4.37\text{A}$$

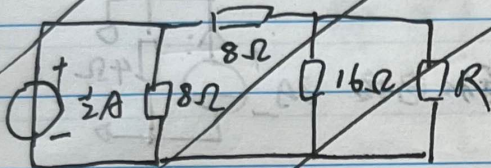
3.16 叠加定理

1. 电压源产生 $I_h = \frac{R+16\Omega}{16\Omega+R+16\Omega} \cdot \frac{U}{12} = \frac{16}{12+16} = \frac{2}{5} A$

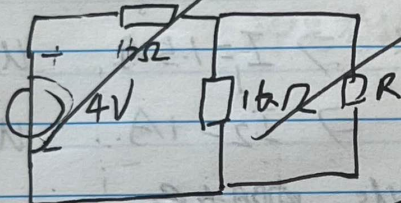
对 g 节点



即



即



由特性定理

$I_g = \frac{1}{5} \cdot \frac{1}{2} A$

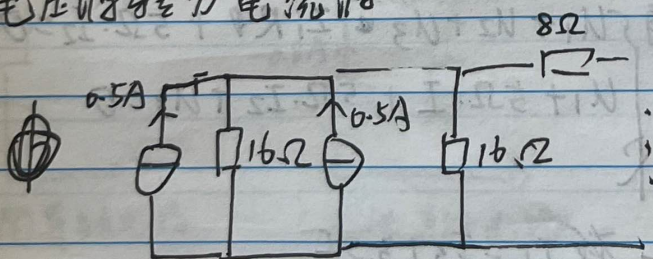
同理，最后有 $I = I_h + I_g + I_f + I_e + I_d + I_c$

$\Rightarrow I = \frac{2}{5} (1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}) = \frac{63}{160} A$

同理 $I_d = I_e = I_f$ $I_f = \frac{1}{5} \cdot 4 A$ $I_e = \frac{1}{5} \cdot \frac{1}{8} A$ $I_d = \frac{1}{5} \cdot \frac{1}{16} A$

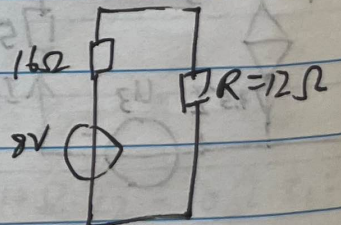
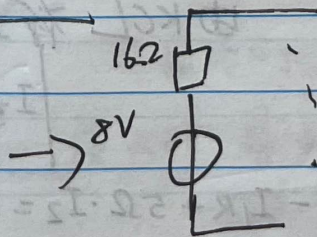
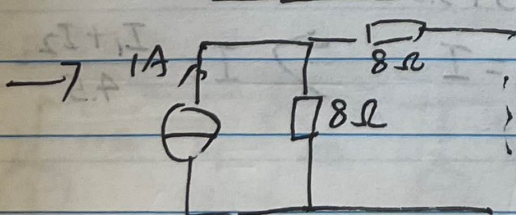
3.16. 电压源转为电流源

即

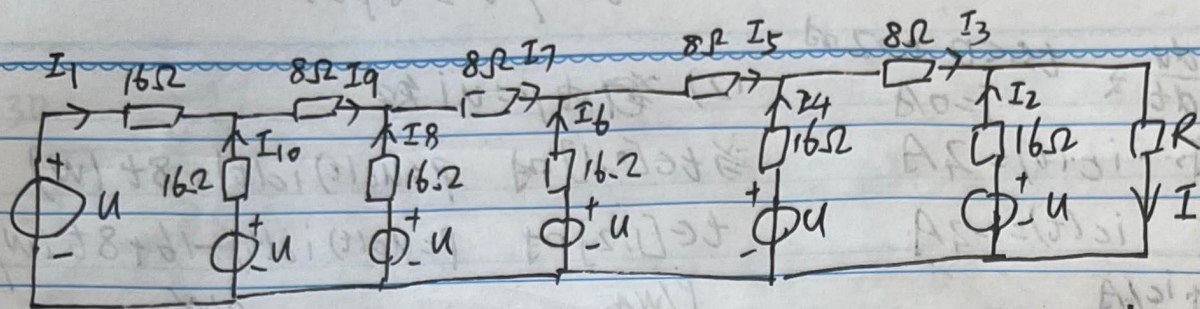


以此类推，由含源网络

电路可等效为



$I = \frac{8V}{16\Omega + R} = 0.4A$

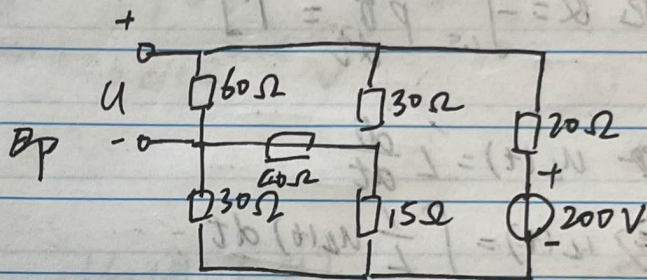
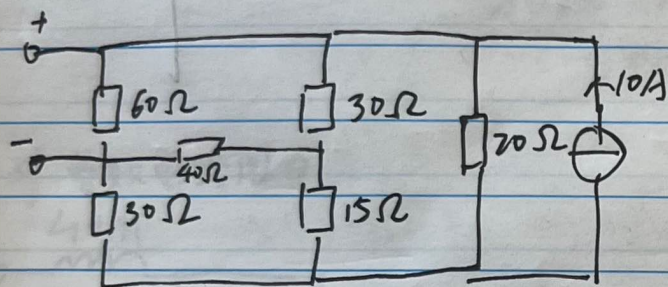


由KVL, $U = I_2 \cdot 16.2 + I \cdot R \Rightarrow I_2 = 0.2A$ 由KCL, $I_3 = 0.2A$

由KVL, $U = I_4 \cdot 16\Omega + I_3 \cdot 8\Omega + I \cdot R \Rightarrow I_4 = 0.1A$ 由KCL, $I_5 = 0.1A$

同理 $I_7 = 0.05A$ $I_9 = 0.025A$ $I_1 = 0.0125A$

3.20 由互易定理, 等效为



显然 40Ω 电阻电桥上无电流

$\therefore \frac{60\Omega}{30\Omega} = \frac{30\Omega}{15\Omega}$

$\therefore R_{\text{外}} = \frac{90\Omega \cdot 45\Omega}{90\Omega + 45\Omega} = 30\Omega$

$\therefore U_{\text{并}} = \frac{30\Omega}{20\Omega + 30\Omega} \cdot 200V = 120V$

$U = \frac{60\Omega}{60\Omega + 30\Omega} \cdot U_{\text{并}} = 80V$